Luiz Takahashi

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Research interests. Measuring cognition and behavior in daily life with wearables and in-home sensors (smartwatch speech, repeated cognitive tests, ambient sensing) for clinical monitoring: when a recorded signal can be trusted, and when a change is real rather than practice, context or noise. Methods: uncertainty-aware machine learning, hierarchical models, calibrated prediction intervals, and validation on people held out of development.

Education

Washington State University, Pullman, WA. B.S. in Computer Science Expected May 2027
GPA 3.83/4.00. Coursework includes Machine Learning, Data Mining, Algorithms and Linear Algebra.

Research Experience

Undergraduate Research Assistant, Washington State University Aug 2025 – Present
Neuropsychology and Aging Lab (Maureen Schmitter-Edgecombe) and CASAS Smart Home Lab (Diane J. Cook). NIH/NIA-funded.

- **ARTS: automated scoring of spoken word-list recall on a smartwatch (first author; submitted to *JINS*).** Adults 50 and older said the 12 words they recalled into an Apple Watch, unsupervised, in daily life (Cognition, Aging and Temperature [CAT] study: 3,443 recordings, 111 participants, 14 days). Designed and built the open-source scoring pipeline on Whisper transcripts (no fine-tuning). It scores from the recognizer's word probabilities, not only its top guess: *candidate rescue* credits a list word that Whisper ranked as an alternative when the word it wrote was off-list and uncertain, and low-confidence words are flagged for human review.
- **Held-out validation, rules frozen first.** Developed the rules on the 91 development participants (602 hand-scored recordings from 14 of them) and froze them. Wrote and coordinated the blind double-transcription protocol that research assistants followed for the held-out set, then evaluated the frozen pipeline on 585 recordings from 20 held-out participants. Correct recall: ICC(2,1) .971 and .968 against two human scorings by different research assistants, which agreed with each other at .987; mean absolute difference 0.21–0.22 words. Repetitions and intrusions (ICC below .20) still need human review.
- **Practice as Signal: practice effects in repeated smartwatch tests (lead; with Cook and Schmitter-Edgecombe).** A test repeated on a watch is also a learning task: in a CAT pilot (113 adults 50 and older), 1-back scores rose 23% in two weeks. Built Praxis, a hierarchical practice-curve model that gives each 1-back prompt a calibrated expected-score band, and tested it on people left out of the fit in two WSU studies (180 adults 50 and older over two weeks; 39 adults in up to five week-long bursts over about 16 months).
- **Leave-one-person-out results.** In the two-week study, a fixed day-1 normal range fell from 93.9% to 42.7% coverage, while Praxis 95% bands held 93–97%. A curve fitted only on the two-week study predicted the 16-month study's first-to-last-burst rise (+4.28 predicted, +4.55 observed).
- **Smart homes (ISG 2026 poster; manuscript in preparation).** Adapted a Kalman filter, with a room-aware mechanism, to event-driven ambient-sensor updates. On synthetic benchmarks generated from the WSU Functional Assessment Study (41 source participants; 30 healthy and 30 impaired simulated residents), it cut localization error by 18% against baseline models and reached 92% statistical consistency in its uncertainty estimates (65% for baselines), and LASSO-ranked behavioral features separated healthy from impaired residents at AUC .78. In 35 older adults' homes (2M+ sensor events), higher 1-back accuracy went with more varied days ($r = -.43$ with day-to-day similarity, $n = 32$, $p = .015$).
- **Medication adherence from in-home sensors (second author, with R. Jenkins; ISG 2026).** Built the medication-event classifier for motion sensors (Metasensors) on pill bottles and organizers: a decision tree trained on 80 clinician-logged events, applied to one week of sensor data from 25 older adults' homes, matched clinician records about 93% of the time. Fitted the adherence models with R. Jenkins.
- **Lab tools.** Extended and evaluated a Presidio-based PHI-redaction tool (EMMA study); worked on scoring tools for D-KEFS verbal-fluency clustering and switching (CAT study). CITI human-subjects training, Oct 2025.

Research Collaborator (remote) with Dr. Junhao Wen, Columbia University 2025 – Present

- Proofs and analysis code for detecting interactions between biomarker views; in multi-cohort proteomic data, an apparent cross-organ interaction vanished under cohort adjustment (an assay-batch Simpson's paradox, as predicted).

Research Assistant, Center for Artificial Intelligence (C4AI), University of São Paulo, Brazil May – Aug 2025

- HarpIA project (Dr. Fabio Cozman). Built SQL Component Match, an AST-based metric that judges whether a generated SQL query matches the reference in meaning: 95% semantic accuracy vs. 70% for the Spider baseline metric; 234-method test suite and a 31-page technical report.

Publications and Other Research Contributions

* marks my name. All authors listed in publication order.

Published conference abstracts

- **L. Takahashi***, D. J. Cook, M. Schmitter-Edgecombe. Room-aware behavioral profiling for cognitive health monitoring in smart homes. *Gerontechnology* 25(Suppl.), p. 1, 2026. doi:10.4017/gt.2026.25.2.1421.3. **Poster**, 15th World Conference of the International Society for Gerontechnology (ISG 2026), Vancouver, Canada, Mar 26–29, 2026. Role: first author; built the models and wrote the abstract.
- R. Jenkins, **L. Takahashi Dos Reis***, D. J. Cook, M. Schmitter-Edgecombe. Remote technological assessment of medication adherence in older adults. *Gerontechnology* 25(Suppl.), p. 1, 2026. doi:10.4017/gt.2026.25.2.1527.3. ISG 2026, Vancouver, Canada. Role: second author; built the medication-event classifier.

Items submitted to peer-reviewed journals

- **L. Takahashi***, A. Adam, H. McDougle, M. Jaworski III, B. Minor, M. Calamia, D. J. Cook, M. Schmitter-Edgecombe. Automated scoring of verbal recall in smartwatch-based ecological momentary assessment: development and independent validation. *Journal of the International Neuropsychological Society*, special issue on AI in neuropsychology. Submitted Sep 2026; decision pending. Role: first author; designed the scoring pipeline and the held-out validation, and wrote the manuscript (M. Schmitter-Edgecombe corresponding; data and human scores from the lab team).

Open-source software

- **ARTS** (github.com/WSU-CASAS/ARTS): scoring pipeline for spoken word-list recall on smartwatches, released with the *JINS* submission. Role: designed and wrote the pipeline, the frozen scoring rules and the reproduction scripts.

Other relevant contributions: manuscripts in preparation

- **L. Takahashi***, D. J. Cook, M. Schmitter-Edgecombe. Practice as Signal: practice-aware expected-score bands for repeated smartwatch cognitive tests (working title). Role: lead; built the model and ran the analyses.
- **L. Takahashi***, M. Schmitter-Edgecombe, D. J. Cook. Coupled to time, stuck in space: dissociable behavioral dimensions in MCI and dementia. Role: first author; analysis and manuscript draft.
- **L. Takahashi***, J. Wen. A completeness hierarchy for multi-view interaction detection, and why naive multi-cohort application fails. Role: first author; proofs and analysis code.

Teaching, Mentoring and Service

- **Teaching Assistant, School of EECS, WSU** (Aug 2025 – Present). Program Design and Development (C/C++): weekly 3-hour lab for 15+ students and office hours on data structures, algorithms and debugging. Data Mining: sole TA for 50+ students, grading assignments and exams.
- **Undergraduate Research Peer Mentor, Office of Undergraduate Research, WSU** (Aug 2026 – Present). Advise students entering research (finding a faculty mentor, paid positions, scholarships, SURCA abstracts) in one-on-one sessions.
- **Volunteer Intern, International Center, WSU** (2025 – Present). Event Programming Committee, 2025–26.

Work Experience

Computer Science Student Worker, Finance and Operations, WSU Alumni Association Sep 2026 – Present

- Rebuilt the monthly reconciliation as a Power Query and Power Automate pipeline with a human approval gate: hands-on matching from about 75 minutes to under 10 per cycle, and staff review from 300+ transactions to 5–15 exceptions. Deployed a locally hosted AI assistant that drafts summaries for staff approval; trained 3 non-technical staff to run it.

Honors and Awards

- President's Honor Roll, Washington State University, all 6 semesters to date. Institutional; academic; non-monetary.
- International Academic Award, Washington State University, 6 semesters, US$2,000 per semester (US$12,000 total). Institutional; academic.
- Scholarships, School of EECS, Voiland College of Engineering and Architecture, WSU: Robert E. Lord Excellence Fund (2026–27, US$4,000); Eunice Hannah Computer Science Scholarship (2025–26, US$3,000); John W. and Anna M. Scott Fund for Academic Excellence (2024–25, US$2,500). Institutional; academic.

Technical Skills

Programming: Python (advanced), C, C++, SQL, Bash, LaTeX; pandas, NumPy, SciPy, PyTorch, scikit-learn, Git, Docker, Linux. **Statistics and ML:** mixed-effects and hierarchical models, split-conformal prediction intervals, calibration, ICC agreement studies, leave-one-person-out and held-out validation, Kalman/UKF state-space filtering, time-series analysis, LASSO. **Speech and sensing:** Whisper ASR (word timestamps and probabilities), smartwatch EMA, ambient smart-home sensors, PHI redaction (Presidio).